



HALDIA INSTITUTE OF TECHNOLOGY

(AN AUTONOMOUS INSTITUTION UNDER MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL)

Paper Code : PCC-CS503

Paper Name : Machine Learning

Time Allotted : 3 Hours

Full Marks :70

The figures in the margin indicate full marks

Candidates are required to give their answers in their own words as far as practicable

Group-A
(Multiple Choice Type Questions)

1. Choose the correct alternatives from the followings:

[15 x 1 = 15]

- | | |
|---|------|
| (I) PCA is | CO-1 |
| (a) Forward feature selection | |
| (b) Backward feature selection | |
| (c) Feature extraction | |
| (d) None of the above | |
| (II) Wheat crops badly damaged on account of rains is | CO-4 |
| (a) Cyclical movement | |
| (b) Random movement | |
| (c) Secular trend | |
| (d) Seasonal movement | |
| (III) The multiple coefficient of determination is calculated as follows: | CO-2 |
| (a) dividing SSR by SST | |
| (b) dividing SST by SSR | |
| (c) dividing SST by SSE | |
| (d) none of the above | |
| (IV) Bagging comes under which method? | CO-3 |
| (a) Sequential methods | |
| (b) Parallel method | |
| (c) Line method | |
| (d) Dot method | |
| (V) The problem of finding hidden structure in unlabeled data is called ____ | CO-2 |
| (a) Supervised learning | |
| (b) Unsupervised learning | |
| (c) Reinforcement learning | |
| (d) None of the above | |
| (VI) Concerning classification algorithms, what is a characteristic of k-Nearest Neighbors? | CO-1 |
| (a) It stores all available cases and classifies new cases based on a similarity measure. | |
| (b) It has been used in statistical estimation and pattern recognition. | |
| (c) It cannot be used for regression. | |
| (d) The input consists of the k closest training examples in the feature space. | |
| (VII) In time series seasonal variations can occur within a period of | CO-4 |
| (a) Four years | |
| (b) Three years | |
| (c) One year | |
| (d) Nine years | |
| (VIII) What is the disadvantage of decision trees? | CO-1 |
| (a) Factor analysis | |
| (b) Decision trees are robust to outliers | |

- (c) Decision trees are prone to be overfit
(d) All of the above
- (IX) What is the minimum number of features required to perform clustering? CO-2
(a) 0
(b) 1
(c) 2
(d) 3
- (X) Which of the following techniques would perform better for reducing dimensions of a dataset? CO-2
(a) Removing columns which have high variance in data
(b) Removing columns which have too many missing value
(c) Removing columns with dissimilar data trends
(d) None of the above
- (XI) Logistic regression is a ____ regression technique that is used to model data having a ____ outcome. CO-1
(a) Linear, binary
(b) Linear, numeric
(c) Nonlinear, binary
(d) Nonlinear, numeric
- (XII) In backpropagation rule, how to stop the learning process? CO-3
(a) No heuristic criteria exist
(b) On basis of average gradient value
(c) There is convergence involved
(d) None of these
- (XIII) Logistic regression is a ____ regression technique that is used to model data having a ____ outcome. CO-1
(a) Linear, binary
(b) Linear, numeric
(c) Nonlinear, binary
(d) Nonlinear, numeric
- (XIV) Which of the following is a subset of machine learning? CO-4
(a) Numpy
(b) Deep Learning
(c) Scipy
(d) All of the above
- (XV) ____ uses ensembles to reduce the variability of single ML models. CO-3
(a) Bagging
(b) Boosting
(c) Stacking/Blending
(d) None of the above

Group-B

(Short Answer Type Questions)

[3 x 5 = 15]

- 2(I) (a) Compare between K-means clustering and K-nearest neighbors algorithm. [2] CO-1
(b) What are the different types of unsupervised learning? [3]

OR

- 2(II) Explain the terms Information Gain and Gini index with the help of an example. [5] CO-1
3(I) (a) What do you mean by Ensemble learning? [2] CO-2
(b) What are the advantages and disadvantages of ensemble learning? [3]

OR

- 3(II) Differentiate between bagging and boosting. [5] CO-2
4(I) Explain what is boosting? [5] CO-3

OR

4(II) What are Feed Forward Networks? Explain.

[5] CO-3

Group-C
(Long Answer Type Questions)

[4 x 10 = 40]

5(I) (a) Write down the steps of KNN algorithm.

[4] CO-1

(b) Following training data is of a group of athletes. Based on this data, use knn algorithm and classify Sayan (Weight = 56kg., Speed=10 kmph) as a Good, Average or Poor sprinter.

[6]

Name	Weight (kg.)	Speed (kmph)	Class
Nitesh	55	9	Average
Gurpreet	58	8	Poor
Goutam	60	7.5	Poor
Gulshan	59	8.5	Average
Mohit	57	10	Good
Sahil	53	10.5	Good
Somak	53	10	Good

OR

5(II) (a) Define Bayes theorem. What are the basic assumptions in Naïve Bayes algorithm?

[3] CO-1

(b) Predict a class label using naïve Bayes classification, given this sample training data. The tuple you have to classify is X= (age=youth, income=medium, student=yes, credit=fair)

[7]

Sl. No	Age	Income	Student	Credit	Buy
1	<30	High	No	Fair	No
2	<30	High	No	Excellent	No
3	31-40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No
7	31-40	Low	Yes	Excellent	Yes
8	<30	Medium	No	Fair	No
9	<30	Low	Yes	Fair	Yes
10	>40	Medium	Yes	Fair	Yes
11	<30	Medium	Yes	Excellent	Yes
12	31-40	Medium	No	Excellent	Yes
13	31-40	High	Yes	Fair	Yes
14	>40	Medium	No	Excellent	No

6(I) (a) Compare between classification and clustering.

[3] CO-2

(b) Apply K-means algorithm using below given for k=2(that is 2 clusters). Use C1(2,2), C2(1,1) as initial cluster centers.

[7]

Data: A(2,2), B(3,2), C(1,1), D(3,1), E(1.5,0.5)

OR

6(II) (a) Explain the working principle of support vector machine algorithm.

[4] CO-2

(b) Find the optimal hyperplane for the set of the data points (1,1),(2,1),(1,-1),(2,-1),(4,0),(5,1),(5,-1),(6,0) and classes. Class 1: {(1,1),(2,1),(1,-1),(2,-1)} and Class 2: {(4,0),(5,1),(5,-1),(6,0)}

[6]

7(I) (a) What do you mean by Ensemble learning?

[5] CO-3

(b) What are the advantages and disadvantages of ensemble learning.

[5]

OR

7(II) (a) What is the general idea of boosting technique?

[5] CO-3

(b) Compare AdaBoost and Gradient tree Boosting.

[5]

8(I) (a) Explain the architecture of an artificial neural network.

[5] CO-4

(b) What is the role of the Activation function in Neural Networks? Explain the various types of activation functions available.

[5]

OR

8(II) (a) What are the limitations of semi-supervised learning?

[5] CO-4

(b) Explain with examples the different components of time series.

[5]

*** END OF PAPER ***